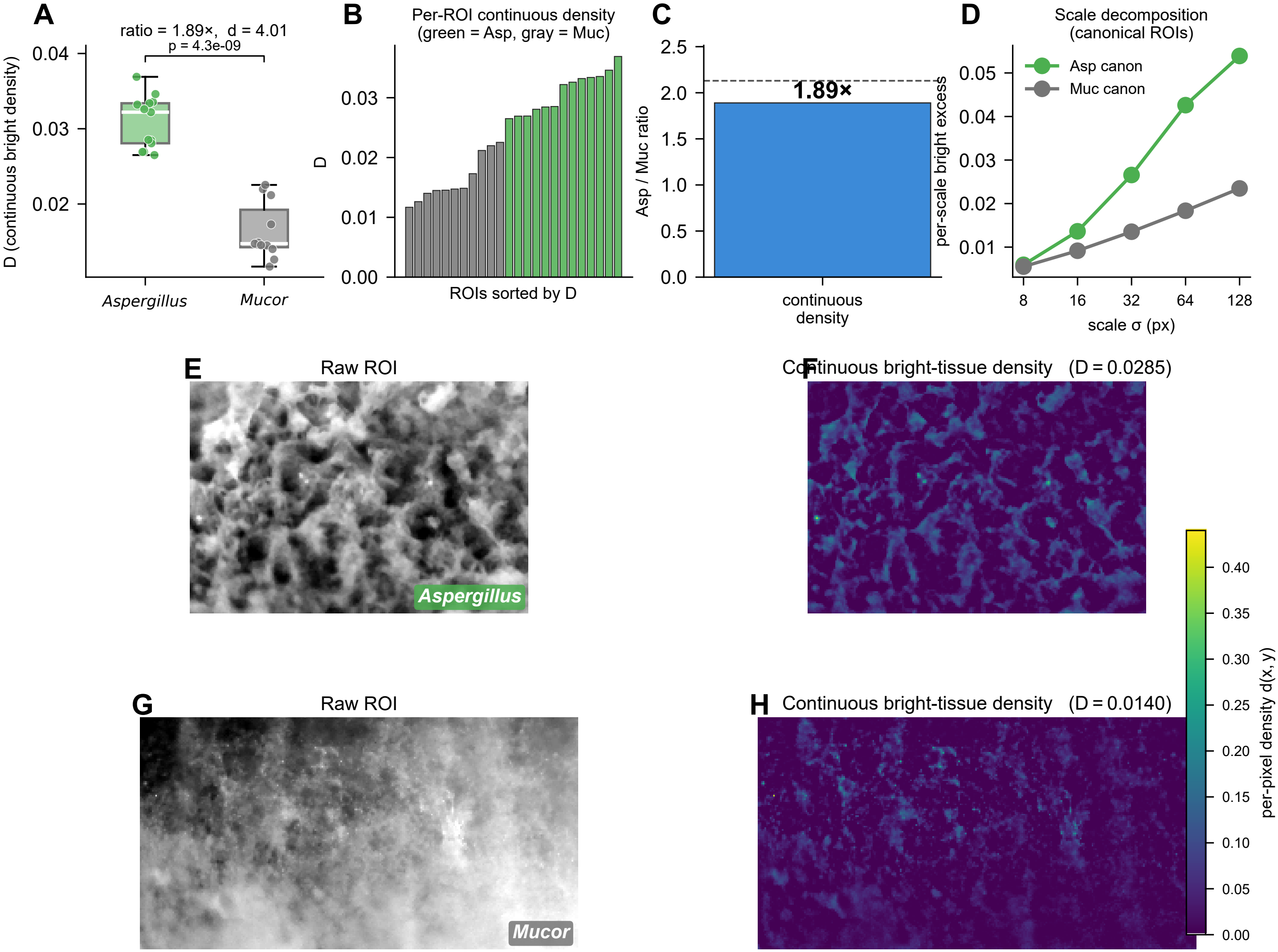

Continuous multi − scale bright − tissue density (this work)

Per-pixel density $d(x,y) = \langle \max(0,\ I_{norm} - gauss(I_{norm}, \sigma))\rangle_{\sigma}$, $\sigma \in \{8, 16, 32, 64, 128\}$ px
Per-ROI scalar $D = \langle d(x,y) \rangle$ over the full ROI

Aspergillus: $D = 0.0309 \pm 0.0035$ ($n = 13$)
Mucor: $D = 0.0164 \pm 0.0038$ ($n = 11$)
Ratio: $1.890\times$, Cohen $d = 4.01$, Welch $p = 4.32e-09$
No overlap? True (δ benchmark = 2.13×)

Compared to binary thresholding: brighter pixels contribute more, grays contribute less, blacks contribute zero — no gating, all the gradient is retained.